

Nguyen Vu Binh Duong

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EDUCATION

University of Engineering and Technology, Vietnam National University (VNU-UET)

Hanoi, Vietnam

M.Sc. in Software Engineering

2023 – 2025

- **GPA:** 3.87/4.00
- **Master's Thesis:** AI-based Software System for Early Diagnosis and Detection of Alzheimer's Disease in Vietnamese People

University of Engineering and Technology, Vietnam National University

Hanoi, Vietnam

B.Sc. in Computer Science

2019 – 2023

- **GPA:** 3.60/4.00
- **Bachelor's Thesis:** Improving the Performance of Unit Test Case Execution for C/C++ Projects

RESEARCH EXPERIENCE

University of Engineering and Technology, Vietnam National University

Hanoi, Vietnam

Lecturer at Faculty of Information Technology

2025 – Present

Lab Instructor/Research Assistant at Faculty of Information Technology

2023 – Present

- Participated in the Alzheimer's Biomarker Cross-border Development (ABCD) Project, Vietnamese-Swiss Joint Research Programme.
 - * Developed a modular machine learning framework, comprising predictor and explainer modules, for Alzheimer's disease (AD) classification using structural MRI data from a Vietnamese cohort (N=113).
 - * Demonstrated that the black-box XGBoost model yielded the maximum classification accuracy (81.4%) relative to other models evaluated on the Vietnamese dataset.
 - * Identified the Generalized Additive Model as the most interpretable and robust approach (78.8% accuracy, 87.2% AUC) for the Vietnamese cohort, balancing performance with clinical explainability.
 - * Validated the hippocampus as a consistent and highly important brain region for AD classification across all models, aligning with established medical findings.
 - * Conducted a comparative analysis of structural features between the Vietnamese cohort (N=76, from April 30th Hospital) and the ADNI cohort (N=80).
 - * Showed that cross-dataset validation resulted in an approximate 10% to 25% drop in prediction accuracy, highlighting the need for population-specific ML models.
- AKAUTAUTO: a proprietary software used for C/C++ unit testing with automated unit test generation as well as classification tree support, GoogleTest integration, multiple self-implemented code coverage types, and many other features.
 - * Provided high-level technical consultation to group members.
 - * Directed the research group and facilitated weekly technical reviews with the joint development team at FPT Software.
- Advanced Programming (INT2215): Delivered lectures on C++ memory management and instructed students on the complexity analysis of algorithms and data structures.
- Object-Oriented Programming (INT2204): Delivered lectures on Object-Oriented Programming principles and mentored students through practical coding with Java, design patterns, and debugging in lab sessions.
- Introduction to Informatics (INT1004 & INT1376): Facilitated lab sessions regarding the Microsoft Office Suite, ensuring students achieved high proficiency in Word, Excel, and PowerPoint.

University of Engineering and Technology, Vietnam National University

Hanoi, Vietnam

Research Student at Software Quality Assurance Lab

2021 – 2023

- Participated in the development of AKAUTAUTO.
 - * Researched and implemented an optimized pairwise test case generation and integrated it with the classification tree feature.
 - * Optimized source code analysis algorithm for C/C++ projects.
 - * Finalized symbolic execution module in the automatic test data generation method of the software.
 - * Implemented and improved stub functions and stub functions with multiple calls feature.
 - * Developed a method to stub functions from built-in and external dependencies.

PUBLICATIONS

Nguyen Tung, L., **Binh Duong, N. V.**, Le, K. N., & Hung, P. N. (2024). *Automated test data generation and stubbing method for C/C++ embedded projects*. *Automated Software Engineering*, 31(2), 52.

Nguyen, M., **Duong, N. V. B.**, Duc, V. T., Phan, C., Phan, T., & Ha, H. (2024, July). *Unveiling Heterogeneity in Early Alzheimer's Disease: A Data-Driven Exploration of Mild Cognitive Impairment and Normal Cognition*. In *International Conference on the Development of Biomedical Engineering in Vietnam* (pp. 408-427). Cham: Springer Nature Switzerland.

Duong Nguyen Vu Binh, Duy-Thanh Vu, Duy-Cat Can, Hai Dac Nguyen, Binh T. Nguyen, Oliver Y. Chén & Huong Ha. *A Comparison of Machine Learning Methods for Alzheimer's Disease Classification in Vietnamese Patients*. *International Symposium on Information and Communication Technology (SOICT 2025)*. Accepted.

Duy-Thanh Vu, **Duong Nguyen Vu Binh**, Oliver Y. Chén, Huong Ha & et al. *Bridging Kernel and Feature Spaces: A New Approach to Accurate and Interpretable Alzheimer's Disease Assessment*. In preparation.

PRESENTATIONS

Oral Presentation

- “Unveiling Heterogeneity in Early Alzheimer's Disease: A Data-Driven Exploration of Mild Cognitive Impairment and Normal Cognition.” at 10th International Conference on the Development of Biomedical Engineering in Vietnam
- “A Comparison of Machine Learning Methods for Alzheimer's Disease Classification in Vietnamese Patients” at International Symposium on Information and Communication Technology (SOICT 2025).

Poster Presentation

- “Machine Learning-Based Analysis of Brain Atrophy Patterns in Alzheimer's Disease: A Comparative Study Between ADNI and Vietnamese Cohorts” at International Conference on Knowledge and Data Acquisition and Applications & Asia-Pacific Symposium on Affective Science and Alzheimer's Disease (IC-KDA & ASAD 2025).

AWARDS AND HONORS

IC-KDA Asian Society Against Dementia Conference Travel Grant, Korea	2025
Vingroup Innovation Foundation Master's Scholarship, Vietnam	2024 – 2025
Second prize, The Scientific Research Conference for Undergraduates at VNU-UET, Vietnam	2023

TECHNICAL SKILLS

MRI processing: CAT12, CONN

Programming: Python, JavaScript, C++

Languages: Vietnamese (native), English (IELTS 7.5 and TOEFL IBT 106/120)

REFERENCES

Assoc. Prof. Huong Ha

Chair | Department of Tissue Engineering & Regenerative Medicine, School of Biomedical Engineering, HCMIU
Email: htthuong@hcmiu.edu.vn

Assoc. Prof. Pham Ngoc Hung

Vice Dean | Faculty of Information and Technology, University of Engineering and Technology, Vietnam National University
Email: hungpn@vnu.edu.vn